Code Cell:

```
%pip install pandas numpy matplotlib seaborn scikit-learn nltk streamlit
```

Output:

```
Requirement already satisfied: pandas in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (3.0.3)
Requirement already satisfied: numpy in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (2.4.6)
Requirement already satisfied: matplotlib in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (3.11.1)
Requirement already satisfied: seaborn in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.13.2)
Requirement already satisfied: scikit-learn in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (1.9.0)
Requirement already satisfied: nltk in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (3.10.2)
Requirement already satisfied: streamlit in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (1.61.1)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pandas) (2026.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib)
(4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib)
(1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib)
(26.1)
Requirement already satisfied: pillow>=9 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: scipy>=1.10.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn)
(1.18.0)
Requirement already satisfied: joblib>=1.4.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn)
(2.24.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn)
(3.6.0)
Requirement already satisfied: defusedxml in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from nltk) (0.7.1)
Requirement already satisfied: click in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from nltk) (8.4.0)
Requirement already satisfied: regex>=2021.8.3 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from nltk)
(2026.7.19)
Requirement already satisfied: tqdm in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from nltk) (4.67.3)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
streamlit) (6.2.2)
Requirement already satisfied: blinker<2,>=1.5.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(1.9.0)
Requirement already satisfied: pydeck<1,>=0.8.0b4 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(0.9.3)
Requirement already satisfied: protobuf<8,>=5.26.1 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(7.35.0)
Requirement already satisfied: pyarrow<25,>=7.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(24.0.0)
Requirement already satisfied: requests<3,>=2.27 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(2.33.1)
Requirement already satisfied: tenacity<10,>=8.1.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(9.1.4)
Requirement already satisfied: toml<2,>=0.10.1 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(0.10.2)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
streamlit) (4.15.0)
Requirement already satisfied: starlette<1.4.0,>=0.46.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
streamlit) (1.3.1)
Requirement already satisfied: uvicorn<1,>=0.30.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(0.52.1)
Requirement already satisfied: httptools<1,>=0.6.3 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(0.8.0)
Requirement already satisfied: anyio<5,>=4.0.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(4.13.0)
Requirement already satisfied: python-multipart<1,>=0.0.10 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
streamlit) (0.0.32)
Requirement already satisfied: websockets<17,>=12.0.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
streamlit) (16.1.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from streamlit)
(6.0.0)
Requirement already satisfied: jinja2 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (3.1.6)
Requirement already satisfied: jsonschema>=3.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (4.26.0)
Requirement already satisfied: idna>=2.8 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
anyio<5,>=4.0.0->streamlit) (3.11)
Requirement already satisfied: colorama in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from click->nltk) (0.4.6)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
requests<3,>=2.27->streamlit) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
requests<3,>=2.27->streamlit) (2.6.3)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
requests<3,>=2.27->streamlit) (2026.2.25)
Requirement already satisfied: h11>=0.8 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
uvicorn<1,>=0.30.0->streamlit) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in C:\Users\Bhara\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
jinja2->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (3.0.3)
```

Output:

```
[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Code Cell:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

Code Cell:

```
import os
print(os.getcwd())
print(os.listdir())
```

Output:

```
C:\Users\Bhara\Desktop\mouni\Amazon.capstone
['.ipynb_checkpoints', 'Amazon.csv', 'amazon.r1.ipynb', 'amazon.r1.py', 'app.py', 'df.pkl', 'Readme.md', 'Readme.txt', 'sentiment_model.pkl',
'tfidf_vectorizer.pkl', 'vectorizer.pkl', 'vectors.pkl']
```

Code Cell:

```
** Dataset Check **
** Data Understanding **
```

Code Cell:

```
import pandas as pd

df = pd.read_csv("Amazon.csv", engine="python")
df.head()
```

Output:

```
     Reviewer Name                    Profile Link Country Review Count  \
0       Eugene ath  /users/66e8185ff1598352d6b3701a      US     1 review
1  Daniel ohalloran  /users/5d75e460200c1f6a6373648c      GB    9 reviews
2         p fisher  /users/546cfcf1000064000197b88f      GB   90 reviews
3        Greg Dunn  /users/62c35cdbacc0ea0012ccaffa      AU    5 reviews
4     Sheila Hannah  /users/5ddbe429478d88251550610e      GB    8 reviews

              Review Date                Rating  \
0  2024-09-16T13:44:26.000Z  Rated 1 out of 5 stars
1  2024-09-16T18:26:46.000Z  Rated 1 out of 5 stars
2  2024-09-16T21:47:39.000Z  Rated 1 out of 5 stars
3  2024-09-17T07:15:49.000Z  Rated 1 out of 5 stars
4  2024-09-16T18:37:17.000Z  Rated 1 out of 5 stars

                              Review Title  \
0      A Store That Doesn't Want to Sell Anything
1          Had multiple orders one turned up and…
2                    I informed these reprobates
3  Advertise one price then increase it on website
4            If I could give a lower rate I would

                                  Review Text  Date of Experience
0  I registered on the website, tried to order a ...  September 16, 2024
1  Had multiple orders one turned up and driver h...  September 16, 2024
2  I informed these reprobates that I WOULD NOT B...  September 16, 2024
3  I have bought from Amazon before and no proble...  September 17, 2024
4  If I could give a lower rate I would! I cancel...  September 16, 2024
```

Code Cell:

```
df = pd.read_csv("Amazon.csv", engine="python", on_bad_lines="skip")

print(df.shape)
df.head()
```

Output:

```
(21214, 9)
```

Output:

```
     Reviewer Name                  Profile Link Country Review Count  \
0       Eugene ath  /users/66e8185ff1598352d6b3701a      US     1 review
1  Daniel ohalloran  /users/5d75e460200c1f6a6373648c      GB    9 reviews
2         p fisher  /users/546cfcf1000064000197b88f      GB   90 reviews
3        Greg Dunn  /users/62c35cdbacc0ea0012ccaffa      AU    5 reviews
4    Sheila Hannah  /users/5ddbe429478d88251550610e      GB    8 reviews

              Review Date                Rating  \
0  2024-09-16T13:44:26.000Z  Rated 1 out of 5 stars
1  2024-09-16T18:26:46.000Z  Rated 1 out of 5 stars
2  2024-09-16T21:47:39.000Z  Rated 1 out of 5 stars
3  2024-09-17T07:15:49.000Z  Rated 1 out of 5 stars
4  2024-09-16T18:37:17.000Z  Rated 1 out of 5 stars

                                Review Title  \
0       A Store That Doesn't Want to Sell Anything
1          Had multiple orders one turned up and…
2                    I informed these reprobates
3  Advertise one price then increase it on website
4            If I could give a lower rate I would

                                         Review Text  Date of Experience
0  I registered on the website, tried to order a ...  September 16, 2024
1  Had multiple orders one turned up and driver h...  September 16, 2024
2  I informed these reprobates that I WOULD NOT B...  September 16, 2024
3  I have bought from Amazon before and no proble...  September 17, 2024
4  If I could give a lower rate I would! I cancel...  September 16, 2024
```

Code Cell:

```
print(df.columns)
```

Output:

```
Index(['Reviewer Name', 'Profile Link', 'Country', 'Review Count',
       'Review Date', 'Rating', 'Review Title', 'Review Text',
       'Date of Experience'],
      dtype='str')
```

Code Cell:

```
df
```

Output:

```
           Reviewer Name                  Profile Link Country  \
0             Eugene ath  /users/66e8185ff1598352d6b3701a      US
1        Daniel ohalloran  /users/5d75e460200c1f6a6373648c      GB
2               p fisher  /users/546cfcf1000064000197b88f      GB
3              Greg Dunn  /users/62c35cdbacc0ea0012ccaffa      AU
4          Sheila Hannah  /users/5ddbe429478d88251550610e      GB
...                  ...                             ...     ...
21209           Anders T  /users/47bd4ffe0000640001001044      DK
21210            David E  /users/495bbbc0000064000100a972      US
21211      Joseph Harding  /users/48cfacbf0000640001005d04      GB
21212         Mads Dørup  /users/474aaec70000640001000a44      DK
21213  Kim Fuglsang Kramer  /users/46d1ed150000640001000051      DK

      Review Count              Review Date                Rating  \
0         1 review  2024-09-16T13:44:26.000Z  Rated 1 out of 5 stars
1        9 reviews  2024-09-16T18:26:46.000Z  Rated 1 out of 5 stars
2       90 reviews  2024-09-16T21:47:39.000Z  Rated 1 out of 5 stars
3        5 reviews  2024-09-17T07:15:49.000Z  Rated 1 out of 5 stars
4        8 reviews  2024-09-16T18:37:17.000Z  Rated 1 out of 5 stars
...            ...                       ...                     ...
21209    1 review  2009-03-22T13:14:12.000Z  Rated 5 out of 5 stars
21210   2 reviews  2008-12-31T18:57:31.000Z  Rated 5 out of 5 stars
21211   3 reviews  2008-09-16T13:05:05.000Z  Rated 3 out of 5 stars
21212  82 reviews  2008-04-28T11:09:05.000Z  Rated 5 out of 5 stars
21213   2 reviews  2007-08-27T17:25:01.000Z  Rated 4 out of 5 stars

                                Review Title  \
0          A Store That Doesn't Want to Sell Anything
1             Had multiple orders one turned up and…
2                       I informed these reprobates
3     Advertise one price then increase it on website
4               If I could give a lower rate I would
...                                              ...
21209                                         Fast!!
21210                          Consistently Excellent
21211          Good prices but delivery can take time :(
21212                    World-class online shopping
21213                                       No title

                                         Review Text  Date of Experience
0     I registered on the website, tried to order a ...  September 16, 2024
1     Had multiple orders one turned up and driver h...  September 16, 2024
```

Code Cell:

```
df = pd.read_csv(
    "Amazon.csv",
    encoding="utf-8",
    engine="python",
    on_bad_lines="skip"
)

print("Dataset loaded successfully")
print(df.head())
print(df.shape)
```

Output:

```
Dataset loaded successfully
      Reviewer Name                    Profile Link Country Review Count  \
0        Eugene ath  /users/66e8185ff1598352d6b3701a      US     1 review
1  Daniel ohalloran  /users/5d75e460200c1f6a6373648c      GB    9 reviews
2          p fisher  /users/546cfcf1000064000197b88f      GB   90 reviews
3         Greg Dunn  /users/62c35cdbacc0ea0012ccaffa      AU    5 reviews
4     Sheila Hannah  /users/5ddbe429478d88251550610e      GB    8 reviews

              Review Date                 Rating  \
0  2024-09-16T13:44:26.000Z  Rated 1 out of 5 stars
1  2024-09-16T18:26:46.000Z  Rated 1 out of 5 stars
2  2024-09-16T21:47:39.000Z  Rated 1 out of 5 stars
3  2024-09-17T07:15:49.000Z  Rated 1 out of 5 stars
4  2024-09-16T18:37:17.000Z  Rated 1 out of 5 stars

                                Review Title  \
0       A Store That Doesn't Want to Sell Anything
1           Had multiple orders one turned up and…
2                    I informed these reprobates
3  Advertise one price then increase it on website
4             If I could give a lower rate I would

                                         Review Text  Date of Experience
0  I registered on the website, tried to order a ...  September 16, 2024
1  Had multiple orders one turned up and driver h...  September 16, 2024
2  I informed these reprobates that I WOULD NOT B...  September 16, 2024
3  I have bought from Amazon before and no proble...  September 17, 2024
4  If I could give a lower rate I would! I cancel...  September 16, 2024
(21214, 9)
```

Code Cell:

```
print(df.info())
print(df.describe())
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 21214 entries, 0 to 21213
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Reviewer Name      21214 non-null  str
 1   Profile Link       21163 non-null  str
 2   Country            21054 non-null  str
 3   Review Count       21055 non-null  str
 4   Review Date        21055 non-null  str
 5   Rating             21055 non-null  str
 6   Review Title       21055 non-null  str
 7   Review Text        21055 non-null  str
 8   Date of Experience  20947 non-null  str
dtypes: str(9)
memory usage: 13.7 MB
None
       Reviewer Name      Profile Link Country Review Count  \
count          21214             21163   21054        21055
unique         18531             21156     148          177
top         customer  January 21, 2012      US     1 review
freq             72                 3    9286         5761

                  Review Date                 Rating Review Title  \
count                   21055                  21055        21055
unique                  21054                      5        19277
top     2022-10-05T12:13:39.000Z  Rated 1 out of 5 stars    Excellent
freq                        2                  13123           52

                Review Text Date of Experience
count                 21055              20947
unique                20407               3640
top     Review text not found   January 11, 2021
freq                    630                 32
```

Code Cell:

```
print(df.isnull().sum())
```

Output:

```
Reviewer Name          0
Profile Link           51
Country               160
Review Count          159
Review Date           159
Rating                159
Review Title          159
Review Text           159
Date of Experience    267
dtype: int64
```

Code Cell:

```
** Duplicate Records Remove **
```

Code Cell:

```
print("Duplicate records:", df.duplicated().sum())
```

Output:

```
Duplicate records: 2
```

Code Cell:

```
df = df.drop_duplicates()

print("Shape after removing duplicates:", df.shape)
```

Output:

```
Shape after removing duplicates: (21212, 9)
```

Code Cell:

```
** Missing Values Handle **
```

Code Cell:

```
print(df.isnull().sum())
```

Output:

```
Reviewer Name          0
Profile Link           49
Country               158
Review Count          157
Review Date           157
Rating                157
Review Title          157
Review Text           157
Date of Experience    265
dtype: int64
```

Code Cell:

```
df = df.dropna()
df
```

Output:

```
           Reviewer Name                  Profile Link Country  \
0            Eugene ath  /users/66e8185ff1598352d6b3701a      US
1      Daniel ohalloran  /users/5d75e460200c1f6a6373648c      GB
2             p fisher  /users/546cfcf1000064000197b88f      GB
3           Greg Dunn  /users/62c35cdbacc0ea0012ccaffa      AU
4         Sheila Hannah  /users/5ddbe429478d88251550610e      GB
...                 ...                             ...     ...
21209          Anders T  /users/47bd4ffe0000640001001044      DK
21210           David E  /users/495bbbc0000064000100a972      US
21211      Joseph Harding  /users/48cfacbf0000640001005d04      GB
21212         Mads Dørup  /users/474aaec70000640001000a44      DK
21213  Kim Fuglsang Kramer  /users/46d1ed150000640001000051      DK

      Review Count             Review Date                Rating  \
0        1 review  2024-09-16T13:44:26.000Z  Rated 1 out of 5 stars
1       9 reviews  2024-09-16T18:26:46.000Z  Rated 1 out of 5 stars
2      90 reviews  2024-09-16T21:47:39.000Z  Rated 1 out of 5 stars
3       5 reviews  2024-09-17T07:15:49.000Z  Rated 1 out of 5 stars
4       8 reviews  2024-09-16T18:37:17.000Z  Rated 1 out of 5 stars
...           ...                       ...                     ...
21209    1 review  2009-03-22T13:14:12.000Z  Rated 5 out of 5 stars
21210   2 reviews  2008-12-31T18:57:31.000Z  Rated 5 out of 5 stars
21211   3 reviews  2008-09-16T13:05:05.000Z  Rated 3 out of 5 stars
21212  82 reviews  2008-04-28T11:09:05.000Z  Rated 5 out of 5 stars
```

Code Cell:

```
print("Columns in dataset:")
print(df.columns.tolist())
```

Output:

```
Columns in dataset:
['Reviewer Name', 'Profile Link', 'Country', 'Review Count', 'Review Date', 'Rating', 'Review Title', 'Review Text', 'Date of Experience']
```

Code Cell:

```
** Review Rating Check **
```

Code Cell:

```
# Remove extra spaces from column names
df.columns = df.columns.str.strip()

# Remove rows where Review Text is missing
df = df.dropna(subset=["Review Text"])

# Fill missing Review Title values
df["Review Title"] = df["Review Title"].fillna("Unknown")

# Check the data
print("Dataset Shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())

print("\nFirst 5 Reviews:")
print(df[["Review Title", "Review Text", "Rating"]].head())
```

Output:

```
Dataset Shape: (20946, 9)

Columns:
['Reviewer Name', 'Profile Link', 'Country', 'Review Count', 'Review Date', 'Rating', 'Review Title', 'Review Text', 'Date of Experience']

First 5 Reviews:
                          Review Title  \
0      A Store That Doesn't Want to Sell Anything
1         Had multiple orders one turned up and…
2                  I informed these reprobates
3  Advertise one price then increase it on website
4           If I could give a lower rate I would

                                 Review Text                 Rating
0  I registered on the website, tried to order a ...  Rated 1 out of 5 stars
1  Had multiple orders one turned up and driver h...  Rated 1 out of 5 stars
2  I informed these reprobates that I WOULD NOT B...  Rated 1 out of 5 stars
3  I have bought from Amazon before and no proble...  Rated 1 out of 5 stars
4  If I could give a lower rate I would! I cancel...  Rated 1 out of 5 stars
```

Code Cell:

```
df = df.dropna(subset=["Review Text"])
df
```

Output:

```
          Reviewer Name                    Profile Link Country  \
0             Eugene ath  /users/66e8185ff1598352d6b3701a      US
1       Daniel ohalloran  /users/5d75e460200c1f6a6373648c      GB
2              p fisher  /users/546cfcf1000064000197b88f      GB
3             Greg Dunn  /users/62c35cdbacc0ea0012ccaffa      AU
4          Sheila Hannah  /users/5ddbe429478d88251550610e      GB
...                  ...                              ...     ...
21209          Anders T  /users/47bd4ffe0000640001001044      DK
21210           David E  /users/495bbbc0000064000100a972      US
21211      Joseph Harding  /users/48cfacbf0000640001005d04      GB
21212         Mads Dørup  /users/474aaec70000640001000a44      DK
21213  Kim Fuglsang Kramer  /users/46d1ed150000640001000051      DK

     Review Count             Review Date                 Rating  \
0        1 review  2024-09-16T13:44:26.000Z  Rated 1 out of 5 stars
1       9 reviews  2024-09-16T18:26:46.000Z  Rated 1 out of 5 stars
2      90 reviews  2024-09-16T21:47:39.000Z  Rated 1 out of 5 stars
3       5 reviews  2024-09-17T07:15:49.000Z  Rated 1 out of 5 stars
4       8 reviews  2024-09-16T18:37:17.000Z  Rated 1 out of 5 stars
...           ...                       ...                     ...
21209    1 review  2009-03-22T13:14:12.000Z  Rated 5 out of 5 stars
21210   2 reviews  2008-12-31T18:57:31.000Z  Rated 5 out of 5 stars
21211   3 reviews  2008-09-16T13:05:05.000Z  Rated 3 out of 5 stars
21212  82 reviews  2008-04-28T11:09:05.000Z  Rated 5 out of 5 stars
21213   2 reviews  2007-08-27T17:25:01.000Z  Rated 4 out of 5 stars

                          Review Title  \
```

Code Cell:

```
# Clean column names
df.columns = df.columns.str.strip()

# Remove rows with missing review text
df = df.dropna(subset=["Review Text"])

# Fill missing review titles
df["Review Title"] = df["Review Title"].fillna("Unknown")

print("Data cleaning completed!")
print("Dataset shape:", df.shape)
print(df.columns.tolist())
```

Output:

```
Data cleaning completed!
Dataset shape: (20946, 9)
['Reviewer Name', 'Profile Link', 'Country', 'Review Count', 'Review Date', 'Rating', 'Review Title', 'Review Text', 'Date of Experience']
```

Code Cell:

```
print(df[["Review Title", "Review Text", "Rating"]].head())
```

Output:

```
                                Review Title  \
0       A Store That Doesn't Want to Sell Anything
1          Had multiple orders one turned up and…
2                    I informed these reprobates
3  Advertise one price then increase it on website
4             If I could give a lower rate I would

                                         Review Text                 Rating
0  I registered on the website, tried to order a ...  Rated 1 out of 5 stars
1  Had multiple orders one turned up and driver h...  Rated 1 out of 5 stars
2  I informed these reprobates that I WOULD NOT B...  Rated 1 out of 5 stars
3  I have bought from Amazon before and no proble...  Rated 1 out of 5 stars
4  If I could give a lower rate I would! I cancel...  Rated 1 out of 5 stars
```

Code Cell:

```
print(df.columns.tolist())
```

Output:

```
['Reviewer Name', 'Profile Link', 'Country', 'Review Count', 'Review Date', 'Rating', 'Review Title', 'Review Text', 'Date of Experience']
```

Code Cell:

```
df.columns = df.columns.str.strip().str.lower()

print(df.columns.tolist())
```

Output:

```
['reviewer name', 'profile link', 'country', 'review count', 'review date', 'rating', 'review title', 'review text', 'date of experience']
```

Code Cell:

```
print("Columns:")
print(df.columns.tolist())

print("\nShape:")
print(df.shape)

print("\nFirst 5 rows:")
display(df.head())
```

Output:

```
Columns:
['reviewer name', 'profile link', 'country', 'review count', 'review date', 'rating', 'review title', 'review text', 'date of experience']

Shape:
(20946, 9)

First 5 rows:
```

Code Cell:

```
print(df["rating"].value_counts())
```

Output:

```
rating
Rated 1 out of 5 stars    13120
Rated 5 out of 5 stars     4447
Rated 4 out of 5 stars     1278
Rated 3 out of 5 stars     1227
```

Code Cell:

```
df["rating_number"] = df["rating"].str.extract(r"(\d+)").astype(int)

print(df["rating_number"].value_counts().sort_index())
```

Output:

```
rating_number
1    13120
2     1227
3      874
4     1278
5     4447
Name: count, dtype: int64
```

Code Cell:

```
** Sentiment Create **
```

Code Cell:

```
def sentiment(rating):
    if rating >= 4:
        return "Positive"
    elif rating == 3:
        return "Neutral"
    else:
        return "Negative"

df["sentiment"] = df["rating_number"].apply(sentiment)

print(df["sentiment"].value_counts())
```

Output:

```
sentiment
Negative    14347
Positive     5725
Neutral       874
Name: count, dtype: int64
```

Code Cell:

```
# Check columns
print(df.columns.tolist())

# Extract numeric rating
df["rating_number"] = df["rating"].str.extract(r"(\d+)").astype(int)

# Check rating distribution
print(df["rating_number"].value_counts().sort_index())

# Create sentiment
def sentiment(rating):
    if rating >= 4:
        return "Positive"
    elif rating == 3:
        return "Neutral"
    else:
        return "Negative"

df["sentiment"] = df["rating_number"].apply(sentiment)

# Check sentiment distribution
print(df["sentiment"].value_counts())
```

Output:

```
['reviewer name', 'profile link', 'country', 'review count', 'review date', 'rating', 'review title', 'review text', 'date of experience', 'rating_number', 'sentiment']
rating_number
1    13120
2     1227
3      874
4     1278
5     4447
Name: count, dtype: int64
sentiment
Negative    14347
Positive     5725
Neutral       874
Name: count, dtype: int64
```

Code Cell:

```
def sentiment(rating):
    if rating >= 4:
        return "Positive"
    elif rating == 3:
        return "Neutral"
    else:
        return "Negative"

df["Sentiment"] = df["rating_number"].apply(sentiment)

print(df["Sentiment"].value_counts())
```

Output:

```
Sentiment
Negative    14347
Positive     5725
Neutral       874
Name: count, dtype: int64
```

Code Cell:

```
** ReviewLength Create **
```

Code Cell:

```
df["ReviewLength"] = df["review text"].astype(str).apply(
    lambda x: len(x.split())
)

print(df[["review text", "ReviewLength"]].head())
```

Output:

```
                                  review text  ReviewLength
0  I registered on the website, tried to order a ...           106
1  Had multiple orders one turned up and driver h...            53
2  I informed these reprobates that I WOULD NOT B...           122
3  I have bought from Amazon before and no proble...            82
4  If I could give a lower rate I would! I cancel...           100
```

Code Cell:

```
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength']
```

Code Cell:

```
** Year and Month Extract **
```

Code Cell:

```
df["review_date"] = pd.to_datetime(
    df["review_date"],
    errors="coerce"
)
print(df["review_date"].head())
```

Output:

```
0   2024-09-16 13:44:26+00:00
1   2024-09-16 18:26:46+00:00
2   2024-09-16 21:47:39+00:00
3   2024-09-17 07:15:49+00:00
4   2024-09-16 18:37:17+00:00
Name: review_date, dtype: datetime64[us, UTC]
```

Code Cell:

```
df["Year"] = df["review_date"].dt.year
df["Year"]
```

Output:

```
0       2024
1       2024
2       2024
3       2024
4       2024
        ...
21209    2009
21210    2008
21211    2008
21212    2008
21213    2007
Name: Year, Length: 20946, dtype: int32
```

Code Cell:

```
df["Month"] = df["review_date"].dt.month
df["Month"]
```

Output:

```
0        9
1        9
2        9
3        9
4        9
        ..
21209     3
21210    12
21211     9
21212     4
21213     8
Name: Month, Length: 20946, dtype: int32
```

Code Cell:

```
** NLP Text Preprocessing **
```

Code Cell:

```
** NLTK Install **
```

Code Cell:

```
import nltk

nltk.download("stopwords")
nltk.download("wordnet")
nltk.download("omw-1.4")
```

Output:

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Bhara\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\Bhara\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data]     C:\Users\Bhara\AppData\Roaming\nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
```

Output:

```
True
```

Code Cell:

```
** Text Cleaning Function **
```

Code Cell:

```
import re
import string
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

stop_words = set(stopwords.words("english"))
lemmatizer = WordNetLemmatizer()

def preprocess_text(text):
    text = str(text).lower()

    # Remove punctuation
    text = text.translate(str.maketrans("", "", string.punctuation))

    # Tokenization
    words = text.split()

    # Remove stopwords and lemmatize
```

Code Cell:

```
df["Clean Review"] = df["review_text"].apply(preprocess_text)
```

Code Cell:

```
print(df[["review_text", "Clean Review"]].head())
```

Output:

```
                                         review_text  \
0  I registered on the website, tried to order a ...
1  Had multiple orders one turned up and driver h...
2  I informed these reprobates that I WOULD NOT B...
3  I have bought from Amazon before and no proble...
4  If I could give a lower rate I would! I cancel...

                                        Clean Review
0  registered website tried order laptop entered ...
1  multiple order one turned driver phone door nu...
2  informed reprobate would going visit sick rela...
3  bought amazon problem happy service price amaz...
4  could give lower rate would cancelled amazon p...
```

Code Cell:

```
** Exploratory Data Analysis (EDA)  **
```

Code Cell:

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))

sns.countplot(data=df, x="rating")

plt.title("Rating Distribution")
plt.xlabel("Rating")
plt.ylabel("Number of Reviews")
plt.show()
```

Code Cell:

```
print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'Year', 'Month', 'Clean Review']
```

Code Cell:

```
df.columns = df.columns.str.strip()

print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'Year', 'Month', 'Clean Review']
```

Code Cell:

```
print(df.columns[df.columns.duplicated()])
```

Output:

```
Index([], dtype='str')
```

Code Cell:

```
df = df.loc[:, ~df.columns.duplicated()]
```

Code Cell:

```
print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'Year', 'Month', 'Clean Review']
```

Code Cell:

```
** Sentiment Distribution **
```

Code Cell:

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))

sns.countplot(data=df, x="sentiment")

plt.title("Sentiment Distribution")
plt.xlabel("sentiment")
plt.ylabel("Number of Reviews")

plt.show()
```

Code Cell:

```
print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'Year', 'Month', 'Clean Review']
```

Code Cell:

```
print(df.columns.tolist())
print(df["sentiment"].value_counts())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'Year', 'Month', 'Clean Review']
sentiment
Negative    14347
Positive     5725
Neutral       874
Name: count, dtype: int64
```

Code Cell:

```
print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'Year', 'Month', 'Clean Review']
```

Code Cell:

```
print(df.columns[df.columns.duplicated()])
```

Output:

```
Index([], dtype='str')
```

Code Cell:

```
print("category" in df.columns)
print("Sentiment" in df.columns)
print("sentiment" in df.columns)
```

Output:

```
False
False
True
```

Code Cell:

```
df["category"] = df["rating_number"].map({
    1: "1 Star",
    2: "2 Stars",
    3: "3 Stars",
    4: "4 Stars",
    5: "5 Stars"
})
```

Code Cell:

```
** Sentiment Distribution by Category **
```

Code Cell:

```
plt.figure(figsize=(12,6))

sns.countplot(
    data=df,
    x="category",
    hue="sentiment"
```

Code Cell:

** Top 10 Most Reviewed Products **

Code Cell:

```
top_country = df["country"].value_counts().head(10)

plt.figure(figsize=(10,6))
top_country.sort_values().plot(kind="barh")

plt.title("Top Countries by Reviews")
plt.xlabel("Number of Reviews")
plt.ylabel("Country")
plt.show()
```

Code Cell:

```
df["rating_number"].value_counts().sort_index().plot(kind="bar")
plt.title("Rating Distribution")
plt.title("Top 10 Most Reviewed ")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```

Code Cell:

** Train-Test Split **

Code Cell:

```
from sklearn.model_selection import train_test_split

X = df["Clean Review"]
y = df["sentiment"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

Code Cell:

** TF-IDF Vectorization **

Code Cell:

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    max_features=5000
)

X_train_tfidf = vectorizer.fit_transform(X_train)

X_test_tfidf = vectorizer.transform(X_test)
```

Code Cell:

** Machine Learning Model **

Code Cell:

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)

model.fit(X_train_tfidf, y_train)
```

Output:

```
LogisticRegression(max_iter=1000)
```

Code Cell:

** Prediction **

Code Cell:

```
y_pred = model.predict(X_test_tfidf)

print(y_pred[:10])
```

Output:

```
['Negative' 'Positive' 'Negative' 'Negative' 'Negative' 'Negative'
 'Negative' 'Positive' 'Negative' 'Negative']
```

Code Cell:

** Model Evaluation **

Code Cell:

```
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix
)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

print(classification_report(y_test, y_pred))
```

Output:

```
Accuracy: 0.8894988066825775
              precision    recall  f1-score   support

    Negative       0.91      0.96      0.93      2870
     Neutral       0.67      0.01      0.02       175
    Positive       0.84      0.84      0.84      1145

    accuracy                           0.89      4190
   macro avg       0.81      0.61      0.60      4190
weighted avg       0.88      0.89      0.87      4190
```

Code Cell:

** Confusion Matrix **

Code Cell:

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(7,5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues"
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()
```

Code Cell:

** Save Model **

Code Cell:

```
import joblib

joblib.dump(model, "sentiment_model.pkl")
joblib.dump(vectorizer, "tfidf_vectorizer.pkl")

print("Model and vectorizer saved successfully!")
```

Output:

```
Model and vectorizer saved successfully!
```

Code Cell:

** Model Load **

Code Cell:

```
import joblib

model = joblib.load("sentiment_model.pkl")
vectorizer = joblib.load("tfidf_vectorizer.pkl")

joblib.dump(vectorizer, "vectorizer.pkl")
joblib.dump(vectors, "vectors.pkl")
joblib.dump(df, "df.pkl")

print("I love this product. Battery life is excellent")
```

Output:

```
I love this product. Battery life is excellent
```

Code Cell:

```
review = ["This product is amazing"]

review_vector = vectorizer.transform(review)

prediction = model.predict(review_vector)

print(prediction)
```

Output:

```
['Positive']
```

Code Cell:

```
prediction = model.predict(review_vector)

print("Predicted Sentiment:", prediction[0])

if prediction[0] == "Positive":
    print("\nRecommended Products:")
    print("1. Product A")
    print("2. Product B")
    print("3. Product C")

elif prediction[0] == "Negative":
    print("\nRecommended Products:")
    print("1. Customer Support")
    print("2. Alternative Product")
    print("3. Refund / Replacement")

else:
    print("\nNeutral Review")
```

Output:

```
Predicted Sentiment: Positive

Recommended Products:
1. Product A
2. Product B
3. Product C
```

Code Cell:

```
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'year', 'month', 'clean_review', 'category']
```

Code Cell:

```
import re

def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    text = re.sub(r" ", " ", text).strip()
    return text

df["clean_review"] = df["review_text"].apply(clean_text)
```

Output:

```
                                        review_text  \
0  I registered on the website, tried to order a ...
1  Had multiple orders one turned up and driver h...
2  I informed these reprobates that I WOULD NOT B...
3  I have bought from Amazon before and no proble...
4  If I could give a lower rate I would! I cancel...

                                        clean_review
0  i registered on the website tried to order a l...
1  had multiple orders one turned up and driver h...
2  i informed these reprobates that i would not b...
3  i have bought from amazon before and no proble...
4  if i could give a lower rate i would i cancell...
```

Code Cell:

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    max_features=5000,
    stop_words="english"
)

vectors = vectorizer.fit_transform(df["clean_review"])

print("TF-IDF Shape:", vectors.shape)
```

Output:

```
TF-IDF Shape: (20946, 5000)
```

Code Cell:

```
** Streamlit Application **
```

Code Cell:

```
import streamlit as st
import joblib
import re
import string

from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

from sklearn.metrics.pairwise import cosine_similarity
```

Code Cell:

```
model = joblib.load("sentiment_model.pkl")
vectorizer = joblib.load("tfidf_vectorizer.pkl")

joblib.dump(vectorizer, "vectorizer.pkl")
joblib.dump(vectors, "vectors.pkl")
joblib.dump(df, "df.pkl")
```

Output:

```
['df.pkl']
```

Code Cell:

```
review = input("Enter your product review: ")

print("\nYour review:")
print(review)
```

Output:

```
Enter your product review:  Very Much Excellent
```

Output:

```

Your review:
Very Much Excellent
```

Code Cell:

```
cleaned_review = preprocess_text(review)

review_vector = vectorizer.transform([cleaned_review])

prediction = model.predict(review_vector)[0]

print("Predicted Sentiment:", prediction)
```

Output:

Predicted Sentiment: Negative

Code Cell:

** Predict Sentiment **

Code Cell:

```
def preprocess_text(text):
    text = str(text).lower()

    text = text.translate(
        str.maketrans("", "", string.punctuation)
    )

    words = text.split()

    words = [
        lemmatizer.lemmatize(word)
        for word in words
        if word not in stop_words
    ]

    return " ".join(words)
```

Code Cell:

```
cleaned_review = preprocess_text(review)
vector = vectorizer.transform([cleaned_review])
print(cleaned_review)
prediction = model.predict(vector)[0]
print(vector)
```

Output:

```
best one
<Compressed Sparse Row sparse matrix of dtype 'float64'
        with 1 stored elements and shape (1, 5000)>
  Coords        Values
  (0, 452)      1.0
```

Code Cell:

```
df["cleaned_review"] = df["review_text"].fillna("").astype(str).apply(preprocess_text)
```

Code Cell:

** Top 3 Similar Products **

Code Cell:

```
input_vector = vectorizer.transform([cleaned_review])
```

Code Cell:

```
from sklearn.metrics.pairwise import cosine_similarity

input_vector = vectorizer.transform([cleaned_review])
similarities = cosine_similarity(
    input_vector,
    product_vectors
).flatten()

print(similarities[:5])
```

Output:

```
[0. 0. 0. 0. 0.]
```

Code Cell:

```
print(df.columns.tolist())
```

Output:

```
['reviewer_name', 'profile_link', 'country', 'review_count', 'review_date', 'rating', 'review_title', 'review_text', 'date_of_experience', 'rating_number',
'sentiment', 'reviewlength', 'year', 'month', 'clean_review', 'category', 'cleaned_review']
```

Code Cell:

```
top_indices = similarities.argsort()[-3:][::-1]

print("Top 3 Similar Customer Reviews")
print("=" * 50)
```

Output:

```
Top 3 Similar Customer Reviews
==================================================
Review Title: Amazon's packaging needs to be addressed
Rating: Rated 1 out of 5 stars
Similarity: 1.0
Country: GB
Review: I recently ordered a nice mug on amazon and I was even excited the night prior to the product's arrival. The  package arrived and when unravelled I was shocked to discover that the mug was packed so poorly that it had cracked!
--------------------------------------------------
Review Title: Very trust
Rating: Rated 5 out of 5 stars
Similarity: 1.0
Country: US
Review: fast shipping
--------------------------------------------------
Review Title: I have been a Prime Member for many…
Rating: Rated 1 out of 5 stars
Similarity: 1.0
Country: US
Review: I have been a Prime Member for many years. Their  delivery is getting very bad.  Every time I order using Prime and the day of delivery it is move to a later date.  I pay for prime I should receive it on the date it states it will arrive !
--------------------------------------------------
```

Code Cell:

```
34. Run Streamlit
```

Code Cell:

```
streamlit run app.py
```

Code Cell:

```
http://192.168.0.102:8501/
```

Code Cell: